

A META-ANALYTIC REVIEW OF SUPPLY CHAIN RISK MANAGEMENT: ASSESSING BUFFERING AND BRIDGING STRATEGIES AND FIRM PERFORMANCE

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Supply chain risk management has received considerable attention as firms experience more frequent and severe impact disruptions. We meta-analytically test the Bode et al. (2011) framework of buffering and bridging supply chain risk management strategies to determine their effect on supply chain risk management. We analyze the supply chain risk management literature to find that both buffering and bridging strategies contribute to supply chain risk management. We also address the benefit of supply chain risk management. Results indicate that supply chain risk management provides a strong contribution to overall firm performance. Additionally, we identify cultural differences of these relationships. Although supply chain risk management strategies may be applied universally, their efficacy varies by culture. In conclusion, we identify and provide guidance for future work.

Keywords: *supply chain risk management; meta-analysis; buffering strategies; bridging strategies; firm performance*

INTRODUCTION

Supply chain disruption is any unplanned and unanticipated event that interrupts the flow of materials or services (Craighead et al., 2007; Kleindorfer & Saad, 2005). These disruptions result in significant financial, operational, and relational costs (Hendricks & Singhal, 2005; Ponomarov & Holcomb, 2009; Sodhi, Son, & Tang, 2012; Speier et al., 2011; Wagner & Bode, 2008). Moreover, the frequency and severity of supply chain disruptions has been increasing (Aon Risk Solutions, 2017; Business Continuity Institute, 2017). For example, supply chain visibility and collaboration have resulted in more cyberattacks due to the interconnectedness of technology, networks, and systems. Recently, 'WannaCry' shut down numerous operations worldwide resulting in financial losses as well as halted patient care in the healthcare sector (Bossert, 2017).

Due to the resulting costs associated with supply chain disruption, organizations have turned their

attention to better understanding and managing supply chain risk (Chopra & Sodhi, 2014), with many firms re-evaluating their supply chain networks. Consequently, organizations have developed supply chain risk management (SCRM) to maintain business continuity and, thus, attenuate these costly disruptions (Hohenstein et al., 2015).

Despite growing interest in SCRM, however, firms struggle to effectively manage their supply chain risk (Melnik et al., 2014). SCRM is defined as 'the identification, assessment, treatment, and monitoring of supply chain risks, with the aid of the *internal* implementation of tools, techniques, and strategies and of *external* coordination and collaboration with supply chain members so as to reduce vulnerability and ensure continuity coupled with profitability, leading to competitive advantage' (Fan & Stevenson, 2018, p. 7). Embedded in this definition are two primary reasons for this struggle: internal constraints (Chang,

Ellinger, & Blackhurst, 2015; Jüttner & Maklan, 2011; Knemeyer, Zinn, & Eroglu, 2009) and the external environment (Slobodow, Abdullah, & Babuschak, 2008).

To address internal and external factors, firms employ two strategies: buffering and bridging (Bode et al., 2011; Fennell & Alexander, 1987; Mezner & Nigh, 1995). On the one hand, buffering strategies have an internal focus and maintain business continuity by establishing safeguards to minimize the firm's exposure to risk and uncertainty in the environment (Bode et al., 2011; Fennell & Alexander, 1987). Bridging strategies, on the other hand, have an external focus and maintain business continuity by attempting to influence the environment through boundary-spanning activities (Bode et al., 2011; Fennell & Alexander, 1987; Mezner & Nigh, 1995). The two strategies take divergent approaches to SCRM, so we therefore investigate them separately.

Further, supply chain networks have become increasingly global, yet many research findings are based upon a single region of the globe (Pagell, Katz, & Sheu, 2005). Subsequently, a growing debate ensues regarding the universal application of research findings (Revilla & Sáenz, 2014). To understand SCRM strategies more fully, we need to understand the effect of cultural differences across nations of the world.

Consequently, the primary purpose of this paper is to provide a systematic and quantitative review of SCRM to reveal patterns unobservable in separate studies (Glass, 1976). The subsequent purpose is to help close the gap in understanding how SCRM enables firms to more effectively manage supply chain risk. We synthesize empirical SCRM work to identify the strength of SCRM relationships to guide future research by also identifying relationships with missing moderators and inconsistent constructs. This results in the following research questions: (1) What is the impact of buffering strategies on SCRM? (2) What is the impact of bridging strategies on SCRM? (3) Are there cultural differences in the relationships between buffering strategies and SCRM and bridging strategies and SCRM? (4) What is the impact of SCRM on firm performance?

We contribute to the literature in the following ways. First, we meta-analytically test the Bode et al. (2011) framework by assessing the strength of relationships between buffering strategies and SCRM and bridging strategies and SCRM. We find independent support for both strategies. Second, we examine both buffering strategies' impact on SCRM and bridging strategies' impact on SCRM across cultural differences. We find that the impact of both bridging strategies and buffering strategies varies by culture. Third, we address the relationship between SCRM and firm performance. We find that SCRM contributes to overall

firm performance and should be considered strategic. Last, we provide a research roadmap outlining the most promising avenues for developing and testing more comprehensive SCRM theory.

In the following section, we present our theoretical foundation and hypothesis development. Next, we describe our methods of data collection, coding and analysis. Third, we report our results. Finally, we discuss theoretical contributions, managerial implications, limitations, and future work.

THEORETICAL FOUNDATIONS AND HYPOTHESIS DEVELOPMENT

Theoretical Foundations

We utilize resource dependence theory (RDT; Pfeffer & Salancik, 1978) and information processing theory (IPT; Tushman & Nadler, 1978) as introduced by Bode et al. (2011). The central position of RDT is that firms must acquire and maintain resources creating dependencies (Pfeffer & Salancik, 1978). Therefore, firms seek independence by safeguarding themselves from the environment (Pfeffer & Salancik, 1978). IPT seeks to reduce information processing needs and/or improve information processing (Huber & Daft, 1987; Tushman & Nadler, 1978). Firms, therefore, simplify complex processes with automation, reducing the number of firms in the supply base, narrowing product lines, and designing products for flexible manufacturing (Flynn & Flynn, 1999). They also create boundary-spanning linkages to gather, interpret, and synthesize information to reduce environmental uncertainty (Frohlich & Westbrook, 2001; Tushman & Nadler, 1978). Both lenses are required to explain all aspects of both buffering strategies and bridging strategies, which differentiate their focus and approach via internal and external dimensions (Fennell & Alexander, 1987; Mezner & Nigh, 1995). On the one hand, buffering has an inward focus, creating internal structures that protect and insulate the firm from the environment (Mezner & Nigh, 1995). On the other hand, bridging has an external focus, creating boundary-spanning linkages to the environment. These linkages reduce the firm's environmental uncertainty by investing in external relationships to gather more information (Flynn & Flynn, 1999).

Buffering Strategies

According to IPT, buffering reduces information-processing needs. Similarly, according to RDT, buffering reduces resource dependence on supply chain network partners. We conceptualize buffering strategies utilizing the measurement items developed by Bode et al. (2011). Specifically, they develop the measurement

items independence, protection, and alternatives to describe how firms insulate themselves from their environment.

Firms gain independence from suppliers when they have manufacturing flexibility to adapt to changing demand and disrupted supply. For example, lean systems utilize flexible work cells of dedicated machinery, which can easily switch production among a variety of products (Womack, Jones, & Roos, 1990). The application of lean tools has demonstrated the improved flow of goods and reduced the number of supply chain disruptions (Schonberger, 2001; Womack & Jones, 1997). Manufacturing flexibility from dedicated machinery, narrow product lines, and postponement of final assembly removes complexity and reduces supply chain risk (Yang & Yang, 2010; Zsidisin, 2003).

Protection stems from slack resources, which often are consumed during a disruption. Slack resources are an attempt to absorb internally the impact of external disruptions (Brandon-Jones, Squire, & Van Rossenberg, 2015). Safety stock inventory, excess capacity, and time can buffer the firm from disruption (Bode, Hübner, & Wagner, 2014; Bourgeois, 1981). Protection also helps prevent supply chain disruption. In addition to slack resources, efforts to secure the supply chain network can help protect against disruptions due to contamination, counterfeits (Maruchek et al., 2011), fire (Jüttner, 2005), terrorism (Sheffi, 2001), and technology threats (Smith et al., 2007).

Lastly, alternatives are replacement or substitute resources that can be used to reconfigure the supply chain during a disruption. Alternatives such as certifying backup suppliers and access to multiple modes of transportation enable firms to reconfigure their supply chain networks with other supply sources or transportation routes (Ambulkar, Blackhurst, & Grawe, 2015; Christopher & Lee, 2004; Tse & Tan, 2012). Thus, taken together, we hypothesize:

H1: Buffering strategies are positively associated with SCRM.

Bridging Strategies

According to IPT, bridging creates linkages with the environment, which increases information-processing capabilities. Similarly, according to RDT, bridging reduces the environment's influence over the firm. We conceptualize bridging strategies utilizing the measurement items developed by Bode et al. (2011). They find firms that employ bridging strategies develop close relationships, monitor, cooperate, exchange information, and conduct joint activities with their supply chain network partners.

Close relationships require high levels of trust and a long-term commitment to develop but result in greater information sharing (Dyer & Singh, 1998).

The information shared in close relationships goes beyond tactical supply and demand to include more proprietary knowledge (Spekman, Kamauff, & Myhr, 1998). Close relationships also can influence suppliers to prioritize allocation of limited resources to firms with whom they have developed long-term relationships (Sheffi, 2001).

Advanced notice of risk can be discovered through monitoring supply chain network partners for any changes or trends (Hallikas et al., 2004). Once risks are identified, they can be assessed by probability and impact. This allows the firm to proactively address risk to help avoid disruptions (Blackhurst, Scheibe, & Johnson, 2008). Monitoring also enables detection of current problems requiring expediting and re-prioritizing to be corrected (Ambulkar et al., 2015).

Developing cooperative relationships with supply chain network partners not only provides access to information but also provides advance warning of potential issues as well as a coordinated response through jointly prepared plans (Revilla & Saenz, 2017). Cooperation is the basis for interdependent firms to coordinate work to improve SCRM (Wieland & Wallenburg, 2013). Cooperation results in faster supply chain disruption response and recovery (Ergun et al., 2010) and in supply chain network partners' willingness to endure monitoring and share information (Wieland & Wallenburg, 2013).

The sharing of strategic and transactional information is required for firms to utilize information to increase control of supply chain risks (Brandon-Jones et al., 2014; Fan et al., 2017; Prajogo & Olhager, 2012). In accordance with IPT, sharing strategies reduce the complexity of the environment (Flynn & Flynn, 1999). Firms can accomplish this through several avenues. First, communicating customer demand information in multitier supply chain networks reduces order variance at upstream tiers (Dejonckheere et al., 2004; Lee, Padmanabhan, & Whang, 1997). Second, supply chain network partners' disclosure of real-time inventory positions helps firms improve delivery performance (Zhou & Benton, 2007). Third, information sharing can directly provide early detection of supply chain disruptions. The more quickly relevant information is shared, the lower the severity of the supply chain disruption (Craighead et al., 2007). Lastly, information sharing also enables cooperation to jointly mitigate risks with supply chain network partners (Braunscheidel & Suresh, 2009). In fact, evidence has shown that the more upstream and downstream tiers of supply chain network partners that are connected together the more continuously the materials flow through the supply chain (Frohlich & Westbrook, 2001).

Inherent in the concept of supply chain networks is the implication that SCRM should be conducted with multiple firms (Jüttner, 2005). Joint activities allow

supply chain network partners to participate in decisions and problem solving and not simply comply with the decisions of others (Richey, Adams, & Dalela, 2012). For example, joint planning improves forecasting and replenishment (Barratt, 2004) and prepares supply chain network partners to make quick decisions when responding to a supply chain disruption (Jüttner & Maklan, 2011). Further, sharing risks and rewards with supply chain network partners aligns incentives and improves SCRM (Li et al., 2015). Thus, we hypothesize:

H2: Bridging strategies are positively associated with SCRM.

Culture

Globalization has increased the inclusion of many more countries within the same supply chain networks, yet we have little insight into how cultural differences impact SCRM (Jia & Zsidisin, 2014; Revilla & Sáenz, 2014). Managerial practices and strategy reflect the collective values and practices of a culture (Hofstede, 1991; Metters et al., 2010). Management and leadership, therefore, vary by culture. This provides insight into the structure and functioning of institutions within various cultures. Although more nuanced studies have been conducted (e.g., Hofstede, 1991; House et al., 2004), there are three overarching ways cultures differ regarding societies' influence on management: (1) inequality and relation to authority, (2) conception of self, and (3) ways of dealing with conflict, uncertainty, and ambiguity (Inkeles & Levinson, 1954). Variation in dealing with conflict, uncertainty, and ambiguity will most certainly affect SCRM.

A good example is Jia and Rutherford (2010) who describe orientation, governance, and relationship building differences between Chinese and Western cultures. They suggest these differences contribute risk to buyer-supplier relationships. Consequently, they offer suggestions for cultural adaptation that will improve supply chain network partnership performance. Thus, we hypothesize buffering strategies and bridging strategies likely will differ across cultures.

H3a: The relationship between buffering strategies and SCRM varies by culture.

H3b: The relationship between bridging strategies and SCRM varies by culture.

Firm Performance

As with RDT and IPT, the goal of SCRM is to reduce vulnerability and ensure continuity (Jüttner, 2005; Wieland & Wallenburg, 2012). Without continuity, firms incur a multitude of costs. Not only do firms incur financial costs of supply chain disruption, which

include back orders, lost sales, and reduced shareholder value (Hendricks & Singhal, 2003; Hendricks & Singhal, 2005), but they also incur operational costs, which include idle resources and expedite costs (Hohenstein et al., 2015; Ponomarev & Holcomb, 2009; Sodhi et al., 2012). Further, relational costs associated with brand damage and loss of reputation with suppliers and employees also result (Kleindorfer & Saad, 2005; Speier et al., 2011).

Not all firms, however, strategically pursue SCRM. SCRM is expensive to implement, and the benefits of a prevented or minimized disruption are difficult to quantify. Disruption costs are incurred only when a disruption occurs (Grötsch, Blome, & Schleper, 2013), so accurately measuring the disruption is difficult (Ganguly, Chatterjee, & Rao, 2018). Further, rewarding employees for preventing problems is an uncommon practice for organizations (Repenning & Sterman, 2001). Thus, to incentivize firms to more effectively manage supply chain risk, we must demonstrate an overall value to the firm. In fact, Narasimhan and Talluri (2009) argue that successful risk identification and assessment must first begin with the identification of overall firm objectives and goals. We therefore hypothesize:

H4: SCRM is positively associated with firm performance.

METHOD

Literature reviews are ubiquitous on topics related to SCRM (e.g., Datta, 2017; Fan & Stevenson, 2018; Friday et al., 2018; Kamalahmadi & Parast, 2016; Kochan & Nowicki, 2018; Tukamuhabwa et al., 2015), indicating a need for consensus and theoretical advancement. However, lacking is a meta-analytic review to assess the nature of these relationships. We tackle this important deficiency to quantitatively and systematically synthesize the growing body of SCRM research.

Systematic literature reviews differ from traditional literature reviews in that they are more inclusive, reducing the chance of bias (Glass, 1976; Leuschner, Rogers, & Charvet, 2013). Only by synthesizing the fully accumulated body of work can the quality of that work be measured (Rousseau, Manning, & Denyer, 2008). Meta-analysis is a method of systematic literature review, which synthesizes the results of primary studies to provide more generalizable and accurate results (Hunter & Schmidt, 2004). Most primary studies lack sufficient power to precisely estimate effect sizes (Lipsey & Wilson, 2001). Thus, meta-analysis enables greater generalizability due to larger sample sizes, while improving precision by correcting for sampling and measurement error and computing the mean correlations weighted by sample sizes (Hunter & Schmidt, 2004).

Sample Selection

Sample selection consisted of four distinct steps: broad searches, closer examination for content, data requirements, and redundancies (e.g., Field & Gillett, 2010; Kong, Dirks, & Ferrin, 2014). First, we conducted a database search in Thomson Reuter's Web of Science, collecting articles published through June 2018. The primary criterion for inclusion was that each study must measure SCRM as either an independent variable or dependent variable. Therefore, we used the following search terms: 'supply chain agility,' 'supply chain disruption,' 'supply chain flexibility,' 'supply chain glitch,' 'supply chain disruption mitigation,' 'supply chain recovery,' 'supply chain resilience,' 'supply chain resiliency,' 'supply chain risk,' 'supply chain robustness,' and 'supply chain vulnerability.'

Due to the aim of being inclusive and because of inconsistencies in determining journal quality, all journals from the ABS Academic Journal Guide were included, in addition to the upper echelon journals (Lipsey & Wilson, 2001). This is in accordance with recommendations by methodological experts (Field & Gillett, 2010; Glass, 1976; Lipsey & Wilson, 2001) and is, therefore, common in meta-analysis (e.g., Cao & Lumineau, 2015; Chen, Damanpour, & Reilly, 2010; Golicic & Smith, 2013; Leuschner et al., 2014; Leuschner et al., 2013; Munyon et al., 2015). In addition to the top journal lists compiled by Mackelprang and Nair (2010) and Mackelprang et al. (2014), we included articles from *International Journal of Logistics Research and Application*; *Supply Chain Management: an International Journal*; *International Journal of Business Performance Management*; *International Journal of Operations and Production Management*; *Journal of Business and Industrial Marketing*; and *Management Decision and Technology Analysis and Strategic Management*.

Second, we conducted a dissertation search in ABI Inform to address potential file drawer bias (Mackelprang et al., 2014). Because they are unpublished, we scrutinized each dissertation in two ways. All accepted dissertations were approved by (1) committee members who have published on the top journal lists (Mackelprang & Nair, 2010; Mackelprang et al., 2014) and (2) programs ranked on the empirical Supply Chain Management list (2019). This added dissertations from Clemson University, Michigan State University, the University of Minnesota, and the University of Toledo.

Third, a manual search was conducted. Journal articles referenced by literature reviews relevant to the topic of SCRM were examined for potential additional studies. The compiled search returned an initial sample size of 1,274 studies. Every effort was made to capture all relevant, qualified papers. Any omission is unintentional.

The next steps of the selection process reviewed papers for content. We excluded papers that had no

empirical data on statistical relationships with SCRM. This eliminated simulations, meta-analyses, conceptual frameworks, and literature reviews. This also included removing studies exploring antecedents of SCRM related directly to firm performance, without relationships to SCRM. This process removed 1,220 papers leaving 54 possible studies.

The requirement of an effect size statistic eliminated additional studies. It is the standardization from using consistent statistics, which allows for the meta-analysis, because such statistics are interpretable in a consistent fashion (Borenstein et al., 2011). The pooled estimate then provides greater statistical power than the individual studies. Therefore, data were scrutinized for correlational data or multivariate data convertible to *r*. Studies were required to report a sample size so that effect sizes could be weighted. This eliminated an additional 17 papers.

Lastly, redundant studies also were rejected. When dissertations or conference proceedings were later published in a peer-reviewed journal, the journal paper was retained, and the earlier work eliminated. This resulted in the elimination of four additional papers, resulting in a sample of 33 qualified studies to code. Although a meta-analysis can be calculated on two studies, a minimum of three studies per relationship is generally accepted practice (Kong et al., 2014). After the coding process, some relationships had insufficient replication to be analyzed. This reduced the number of papers used to 26.

To test cultural differences, we code countries by GLOBE cluster. GLOBE groups 62 national cultures into 10 clusters based upon correlations of cultural dimensions and influence on leadership behavior and effectiveness (House et al., 2004). This is important, because GLOBE is not based on geographical boundaries or subjective evaluations of an East versus West classification (Pagell et al., 2005). We do not investigate GLOBE dimensions but only utilize the framework of clusters to investigate the effectiveness of buffering strategies' and bridging strategies' impact on SCRM across cultural differences. We therefore cluster studies together for Confucian Asia (China, South Korea, and Taiwan), Anglo (the United States and the United Kingdom), and Germanic Europe (Germany, Austria, and Switzerland) to determine whether buffering and bridging strategies' effect on SCRM differs across cultures. Papers utilized in our analyses are described in Appendix A, including publication type, sample size, method, level of analysis, theory, and GLOBE cluster.

Coding Process

The second meta-analytic step was to extract the relevant constructs pertaining to the research question and that possessed the necessary data. The

corresponding variables (i.e., the operationalization of the constructs of interest) are the unique measurement of the individual studies. Thus, each construct was coded according to its operationalization. Construct definitions with representative operationalizations are shown in Appendix B.

The first and second authors coded all of the 26 studies to calculate interrater reliability. Through the coding process, the authors deliberated ambiguous items to confirm uniformity. Once the coding was completed, the authors examined the coding for discrepancies or errors among the common coded studies. We then calculated interrater reliability estimates. Discrepancies and errors were discussed and corrected. Of the 1397 potential study inputs coded by the first two authors, a 94.1 percent agreement on study characteristics and a 98.5 percent agreement on study numbers were met, leading to an overall agreement of 97.2 percent between both raters.

Lower-level operationalizations of constructs were aggregated to represent broader, higher-level-related constructs. These broader constructs enable greater generalization than do the context-specific individual studies and generate consensus by finding the overlap across various studies. This aggregation is described as combining 'apples' and 'oranges' to make 'fruit salad' (Hunter & Schmidt, 2004). Combining similar individual operationalizations resulted in 15 lower-level constructs. The lower-level constructs were aggregated to four higher-level constructs: supply chain risk management, buffering strategies, bridging strategies, and firm performance. We outline construct aggregation in Appendix B.

Meta-Analytic Process

Data were first manually analyzed with Excel to create familiarity with the detailed data. Then, scripts of the Hunter and Schmidt (2004) method developed by Wilson (2010) were run in SPSS. These are macros of well-established SPSS syntax files for running meta-analysis procedures (Field & Gillett, 2010). For a study with more than one reported correlation, we computed composite correlation values to prevent double counting (Hunter & Schmidt, 2004). The Hunter and Schmidt (2004) method progressively makes corrections for individual study variances. The correlation statistic r was used due to its wide availability. Sample-weighted averages were made to each individual study. Then, random variation due to sample size was corrected. Next, variances in independent and dependent variables were corrected for measurement error by multiplying an attenuation factor calculated from the construct reliabilities. Missing reliability data were imputed with average reliability (Lipsey & Wilson, 2001). Variances were corrected by subtracting artifact errors. Consistency was thereby gained by

correcting for measurement and sampling errors. The combined and corrected results are more generalizable to the true population.

The Hunter and Schmidt method utilizes these measures to focus on the direction and magnitude of effects. Most studies focus on significance testing, which does not describe the variance. The contribution of a meta-analysis is a means to make sense of what has been accumulated by providing confidence intervals to show the overlap of separate studies. Also, significance testing does not distinguish between variance due to uncorrected artifacts or moderators. Meta-analysis, however, can correct for measurement error and detect missing moderators of relationships.

RESULTS

Hypothesis Testing

We analyzed data available for each hypothesis and available dimensions of the hypothesized constructs. For each meta-analytic relationship, we present the following results: the number of independent samples (k), uncorrected sample-weighted mean effect size (r), the combined total sample size (n), corrected effect size (p), 95 percent confidence intervals using standard error (CI), 80 percent credibility intervals using standard deviation (CV), and the % of artifact-related total variance explained (%ARTV) by the hypothesized relationships (Hunter & Schmidt, 2004).

The rule of thumb is that effect sizes (p) of .1 are considered small, effect sizes of at least .25 are of medium significance, and those greater than or equal to .4 are largely significant (Cohen, 1977, 1988). Confidence intervals (CI) reflect the amount of error or precision of the mean by using the standard error (Schmidt & Hunter, 2014). They are influenced by sampling error and indicate the range of the mean true score. Intervals containing zero are considered to potentially have a true mean effect of zero. Credibility intervals (CV) express the distribution or magnitude of the population variance by using the standard deviation of the corrected correlations (Schmidt & Hunter, 2014). Wider CVs are less reliable and are influenced by random effects (Schmidt & Hunter, 2014). Wide CVs are a basis for detecting the presence of unidentified moderators (Whitener, 1990).

Moderator variables in the meta-analysis were detected by the 75 percent rule of thumb (Schmidt & Hunter, 1977). Error variances explained by artifact correction (%ARTV) less than 75 percent are considered to indicate missing moderators. Explained variances above 75 percent are generally considered free from large moderator effects (Hunter & Schmidt, 2004). All but one of the relationships (i.e., bridging strategies in Germanic Europe) in this study have % ARTV less than 75 percent, indicating that potential

moderators influence most of the relationships. This suggests more complex relationships should be studied in the future, which report greater detail. Results for all analysis are summarized in Table 1

To test hypothesis 1, we evaluated the correlations between buffering strategies and SCRM. Buffering strategies have a moderate, positive effect on SCRM ($\rho = .39$), with a 95 percent confidence interval that does not contain zero (CI = .20 to .46). To test hypothesis 2, the correlations between bridging strategies and SCRM were evaluated. Bridging strategies have a strong, positive effect on SCRM ($\rho = .47$), with a 95 percent confidence interval that does not contain zero (CI = .29 to .50). These results provide support for hypotheses 1 and 2.

In order to test hypothesis 3a and 3b, we examined the effects of cultural differences on the relationships between buffering strategies and bridging strategies and SCRM. We analyzed GLOBE cultural differences between Confucian Asia (China, South Korea, and Taiwan), Anglo (the United States and the United Kingdom), and Germanic Europe (Germany, Austria, and Switzerland) to determine whether buffering and

bridging strategies' effect on SCRM is different across cultures.

In Confucian Asia, we find strong, positive effects for both buffering strategies ($\rho = .50$; 95% CI = .34 to .54) and bridging strategies ($\rho = .60$; 95% CI = .42 to .61), with neither confidence intervals containing zero. These are the strongest effect sizes of the three cultures. In Anglo, we find moderate, positive effects for buffering ($\rho = .22$; 95% CI = .05 to .32). We find strong, positive effects, however, from bridging ($\rho = .47$) with intervals that do not contain zero (95% CI = .29 to .50). The effect size of bridging is more than double that of buffering. These are the most dissimilar effect sizes between buffering and bridging strategies of the three cultures. In Germanic Europe, we find very similar effect sizes of buffering and bridging ($\rho = .43$ and $\rho = .43$) and intervals that do not contain zero (95% CI = .21 to .49 and .25 to .48, respectively). These are the most similar effect sizes between buffering strategies and bridging strategies of the three cultures.

We further conducted a random effects regression analysis and chi-square test for homogeneity to verify

TABLE 1

Direct Effects on Supply Chain Risk Management									
	Direct Effects on Supply Chain Risk Management								
	k	n	r	ρ	Lower 95% CI	Upper 95% CI	Lower 80% CV	Upper 80% CV	% ARTV
H1a Buffering to SCRM	45	8,464	.33	.39	.20	.46	.10	.67	10.9
H1b Bridging to SCRM	51	12,448	.39	.47	.29	.50	.18	.76	7.5
H2 SCRM to Firm Performance	10	2,581	.35	.40	.24	.45	.04	.75	5.0
H3: Country Effects									
Buffering to SCRM - Confucian Asia	4	1,027	.44	.50	.34	.54	.35	.65	18.3
Bridging to SCRM - Confucian Asia	10	2,440	.52	.60	.42	.61	.38	.83	9.0
Buffering to SCRM - Anglo	12	2,524	.19	.22	.05	.32	-.08	.52	9.8
Bridging to SCRM - Anglo	13	3,153	.39	.47	.29	.50	.11	.82	5.0
Buffering to SCRM - Germanic Europe	5	793	.35	.43	.21	.49	.21	.65	19.0
Bridging to SCRM - Germanic Europe	6	1,296	.37	.43	.25	.48	.38	.48	76.2

TABLE 2

Random Effects Regression Analysis and Chi-Square Test for Homogeneity

Random Effects Regression Analysis and Chi Square Test for Homogeneity	Model w/ out Predictors <i>p</i>	Continuous Predictor			Categorical Predictor			Full Model		
		β	df	<i>p</i>	χ^2	df	<i>p</i>	χ^2	df	<i>p</i>
Buffering Strategies to SCRM with Culture Differences	.000	.265	17	.000	6.708	2	.035	18.594	18	.417
Bridging Strategies to SCRM with Culture Differences	.000	.525	25	.000	1.207	2	.547	27.162	26	.401

the group differences (Table 2). We find a poor model fit for both buffering strategies ($p = .417$) and bridging strategies ($p = .401$); therefore, the null hypothesis of homogeneity is rejected. This indicates statistically significant group differences between cultures (De Veaux, Velleman, & Bock, 2011). This provides further support for hypothesis 3a and 3b.

Hypothesis 4 predicts a positive effect of SCRM on firm performance. SCRM has a large, positive effect on firm performance ($p = .40$) with intervals that do not contain zero (95% CI = .24 to .45). Therefore, hypothesis 4 is supported.

DISCUSSION

Our goal was to provide a systematic and quantitative review to enable greater understanding of the strength of SCRM relationships and guide future work. Meta-analysis allows for related items to be examined to determine more accurate and generalizable relationships (Borenstein et al., 2011). We identified the effect of buffering strategies and bridging strategies to help firms more effectively manage supply chain risk. We also find that culture differences shape the impact of SCRM strategies of buffering and bridging. This is important in establishing conclusive evidence that cultural differences are relevant factors prior to investigating their impact (Pagell et al., 2005). Further, we address the issue of SCRM impact on the firm. Previous work has shown that supply chain disruptions are associated with financial, operational, and relational costs. We find that SCRM can be a source of overall firm performance. Although SCRM is expensive and difficult to implement, it can provide benefits that surpass the investment in safeguards and information processing. Overall, our results suggest that firms can improve SCRM and firm performance by employing both buffering and bridging strategies. Therefore, SCRM should be considered strategic.

Theoretical Contributions

This study provides general support for the Bode et al. (2011) framework as well as both RDT and IPT as

theories to help explain how firms utilize buffering strategies and bridging strategies to manage supply chain risk. Consistent with RDT, we find that firms seek to reduce supply chain dependence and influence from the environment. Also consistent with IPT, we find that firms seek to engage and influence their environment.

Table 1 outlines differences across cultures. First, the effects of buffering strategies and bridging strategies within Confucian Asia are stronger than within other cultures. Second, the effects of buffering strategies and bridging strategies in Anglo firms are more dissimilar than in other cultures (i.e., buffering strategies in Anglo firms are weaker). Third, we found strong effect sizes found from bridging strategies in Germanic European firms. The effects of buffering strategies and bridging strategies within Germanic European firms are more similar than in other cultures. In fact, the effect sizes are the same, and the confidence intervals are very comparable.

Managerial Implications

This study also has implications for management. First, we identify distinct buffering strategies and bridging strategies that contribute to SCRM. Firms have limited resources and face an overwhelming amount of information. This study begins to help them select the strategies they should employ to effectively manage their supply chain risks. We enable selection by analyzing the most frequently studied SCRM relationships within a single paper. The effect of bridging strategies is higher than that of buffering. In fact, the effect size of bridging strategies is outside of the confidence interval for buffering strategies. This does not imply that bridging strategies should be uniformly maximized. There must be a strategic reason for the investment of resources and high-level managerial effort required for bridging strategies. Buffering strategies should be pursued in addition to bridging strategies.

Second, we establish the contribution, which SCRM can make to the firm's bottom line. As opposed to thinking of SCRM as a liability, SCRM may be seen to contribute to improved overall firm performance by

maintaining business continuity. SCRM can be a source of competitive advantage. This is important because when firms view SCRM as a cost to be justified, SCRM becomes a burden (Van Der Vegt et al., 2015) as opposed to a benefit.

The strategies of bridging and buffering may not apply universally to other cultures. This suggests firms should tailor SCRM by location. Failure to recognize cultural differences could hinder the implementation of buffering strategies and bridging strategies. Firms with international supply chain networks must train employees how to adapt to expectations of legal obligations (i.e., Anglo), financial incentives (i.e., Germanic Europe), and reciprocity (i.e., Confucian Asia) to employ more effective bridging strategies. They also may want to exploit bridging strategies to better understand cultural differences (e.g., risk sources, markets) and employ more effective buffering strategies. Firms must evaluate their global footprint to determine the correct mix of alternatives, independence, and protection.

Limitations

As is the case for all research efforts, a few limitations are present. First, we are limited by the availability of primary studies. Although we utilized rigorous procedures to include every relevant study, we cannot guarantee that some evidence was not missed. We are confident, however, that it is unlikely the results would change. Further, some measurements lacked the replication necessary to be included in the analysis. This helps identify areas for new research.

Second, we are limited by the availability of reported data. Some studies did not report reliability information. In these cases, we substituted the average reliability. In addition, unidentified potential moderators are present which have not been captured. Reporting these details could explain relationships more precisely. For example, many studies report combined results for mixed industries and multiple countries. This helps guide future work to investigate and report contextual differences.

Recommendations for Future Work

Many directions for future work can be identified from this study, which will further our understanding of SCRM. We must fully develop research to advance SCRM theory and generate impact (Knemeyer & Fawcett, 2015). We do this first by addressing potential missing moderators, which were identified by the relationships with ARTV less than 75 percent. In addition, we address missing constructs, relationship interdependencies, and causal relationships and conclude with study designs.

Missing Moderators. Future work should compare, contrast, and report more industry-specific details. All but two studies aggregated industries together as either

manufacturing or mixed. This is an important area for future research, because different industries have different levels of industry dynamism, tier level, competitive intensity, market concentration, and product complexity. Thus, industry differences could lead to either buffering strategies or bridging strategies being more effective.

An important potential moderator impacting buffering strategies could be industry dynamism. As the pace of industry change increases, alternatives and protection could quickly become obsolete. Alternative suppliers, modes of transportation, and routes may no longer be certified or viable for new component requirements. Similarly, slack resources such as safety stocks and part-specific tooling could become obsolete. We did not find studies measuring or controlling for industry dynamism, but we did find single studies measuring environmental dynamism, market share, Herfindahl–Hirschman Index (market competitiveness), and competitive intensity. Industry dynamism could hinder the effectiveness of buffering strategies, because slack resources are at higher risk of obsolescence. The higher risk would make implementation under those conditions more expensive.

Another important potential moderator impacting buffering strategies and bridging strategies could be supply chain tier level. Upstream tiers of the supply chain such as raw materials or components are more distant from the end customer than the finished good tier of the supply chain. This distance from the end customer results in less information regarding actual demand and leads to greater variance in orders (Lee, 1997). Information exchange in upstream tiers may be dependent upon the ability of their downstream finished goods supply chain network partners to have close relationships, cooperate, and conduct joint activities with downstream customers to obtain end-customer information. Are bridging strategies less effective for upstream supply chain tiers? Does supply chain tier level moderate the effectiveness of different aspects of bridging strategies?

Similarly, supply chain network complexity is an important potential moderator. Firms that frequently outsource may increase exposure to more sources of risk (Revilla & Sáenz, 2014). The more direct suppliers and tiers of supply chain partners in the network, the greater the frequency of disruptions (Bode & Wagner, 2015). This is due to an inability to identify every supply chain network partner, so some risks could not be identified and, thus, buffered against. Additionally, supply chain network complexity can amplify the severity of supply chain disruptions as they move downstream, creating major disruptions. Safety stock may be insufficient to absorb a major supply chain disruption in a complex supply chain network. Does supply chain network complexity moderate the effectiveness of buffering strategies? A thorough and

TABLE 3

Cultural Descriptions

Culture	Relationship Orientation	Regulatory Forces	Outsourcing	Outcomes
Confucian Asia: China, South Korea, and Taiwan Result-driven, encourage Collective: group working together over individual goals.	Reciprocity of favors (guanxi) including government and competitors with less expected from weaker parties. Long term. Invests in teamwork skills (Cai et al., 2010; Park & Luo, 2001; Power et al., 2010)	Relational, negation and compromise (Cai et al., 2010; Handley & Angst, 2015)	Low levels (Ribbink & Grimm, 2014)	Very high buffering (.50) and bridging (.60)
Anglo: United States and United Kingdom Competitive and results oriented. Low collectivism.	Legal and self-interest exerted from power. Short term. Invests in innovation (Cai et al., 2010; Jia & Rutherford, 2010; Power et al., 2010)	Contractual laws and procedures (Cai et al., 2010; Jia & Rutherford, 2010)	High levels and reliant on Chinese firms (Jia & Zsidisin, 2014; Ribbink & Grimm, 2014)	Unbalanced effects. Moderate buffering (.22) and sometimes no effect. High bridging (.47)
Germanic Europe: Germany, Austria, and Switzerland Value competition and aggressiveness and are more result-oriented. Low collectivism.	Financial incentives and worker participation. Importance from training and education. Weak parties supported by regulation. (Lane, 1997; Milling et al., 2001)	Standardization through trade associations (Lane, 1997; Milling et al., 2001)	Common understanding from standards and technical training enables outsourcing with less communication and misunderstanding (Lane, 1997; Skipper et al., 2008)	High and balanced buffering (.43) and bridging (.43)

TABLE 4

Recommendations for Future Research

Influential Factor	Research Question
Causation	Is there causation between bridging and buffering strategies? Which risk sources of supply chain disruptions are best mitigated by buffering or bridging strategies? Is buffering more effective for natural disasters and socio-economic change while bridging is more effective for supply and demand sources? What aspects of supply chain network complexity are the greatest hindrance to SCRM? Which aspects can be bridged, and which should be buffered? Does identification of uncertainties affect employment of either buffering strategies or bridging strategies?
Countries	Future work should compare, contrast, and report more country-specific work considering factors such as infrastructure, rule of law, regulations, intellectual property protections, and culture. Which factors are the most important in explaining country differences? How are perceptions, motivation, and decision making impacted by international supply chain networks? Can some country-level affects be attributed to the tier in the supply chain network?
Industries	Future work should compare, contrast, and report more industry-specific work, considering factors such as competitive intensity, HHI, tier level, product complexity, and industry dynamism. Which factors and levels are most important to SCRM?
Interdependencies	Under what conditions might buffering and bridging strategies be effective or not? What are the interdependencies between specific bridging strategies such as social capital, collaboration, information sharing, and integration? What is the overlap or tension between alternatives, independence and protection? Is the impact of network complexity on SCRM moderated by effective bridging strategies?
Operationalizations	The supply chain field lacks consensus of terms, and authors utilize various measures. Operationalizations should converge to enable comparison. Measures of response time should move away from subjective self-reports (i.e., slow to fast) to more objective measures (i.e., hours, days, weeks).
Performance	Which specific strategies or dimensions have the greatest effect on overall firm performance? How is strategy selection impacted by attributions of supply chain disruption or performance aspirations?
Replication	Lack of replication removed studies from the analysis. Replication of studies is needed not only to synthesize findings for a more accurate assessment of effect size, but also to identify unknown moderators of relationships. Theory development requires replication and not just novelty.
Temporal	What are the cost benefit tradeoffs of various supply chain disruption response times under different conditions? How do capabilities evolve over time? How is SCRM created, changed, or preserved? How do buffering and bridging strategies relate to the SCRM process? How do interactions between societal factors evolve over time? Do economic pressures provide infrastructure, which then changes cultural and regulatory support?

parsimonious framework for measuring supply chain network complexity in future research is horizontal, vertical, and spatial complexity by Bode and Wagner (2015).

Missing Constructs. This study highlights opportunities for future work by identifying constructs where there were not enough studies to conduct a meta-analysis. Examples of missing or understudied constructs

include postponement (Yang & Yang, 2010), product complexity (Novak & Eppinger, 2001), and organizational culture (Braunscheidel & Suresh, 2009). We need to understand the effect of these potential constructs to understand more complex cause-and-effect relationships and interdependencies.

We did not find papers measuring the effect of postponement on supply chain risk management. Postponement is a buffering strategy to manage high-demand uncertainty by delaying the commitment of resources (Manuj & Mentzer, 2008a, 2008b). When demand is uncertain, firms can apply safety stocks to component assemblies, which are cheaper than finished goods, and then employ flexible manufacturing to quickly assemble finished goods as determined by actual sales. Is buffering more effective when it includes postponement? What are the interdependencies of postponement, buffering, and bridging? Is postponement more effective when bridging strategies are ineffective at determining demand? Or is postponement more effective when bridging strategies effectively convey customer preferences in determining common components?

Novak and Eppinger (2001) found that firms with complex products seek independence from suppliers (i.e., buffering strategy) and produce in-house, because it is assumed that internal coordination is more efficient than external coordination (i.e., bridging strategy). We found two studies measuring the relationship of product complexity to supply chain risk management. Product complexity increases the risk (Merschmann & Thonemann, 2011) and impact of disruption (Bode & Macdonald, 2017). Is the assumption that internal coordination is more efficient than external coordination correct under all conditions? Are there specific bridging strategies with a greater effect on supply chain risk management than buffering strategies? It is important to study the effect of these missing or understudied constructs on buffering strategies and bridging strategies in future research.

In addition to cultural influences, stemming from national differences is firm-level differences. We need descriptions of organizational culture and their effects on buffering strategies and bridging strategies. Although we did not find studies measuring organizational culture, we did find a few studies that measured the enabling effect of specific learning (Ellinger et al., 2015) and disruption (Ambulkar et al., 2015; Riley, 2013) orientations and risk propensity (Park, 2011). The competing value framework (Cameron & Quinn, 2011) is a well-established classification of organizational culture, including dimensions of stability, flexibility, and risk taking that could prove enlightening. For example, Braunscheidel and Suresh (2009) find that a flexible approach and external focus enable

external integration with suppliers and customers. Do certain organizational cultures have a causal relationship with buffering and bridging strategies?

Complex Relationships. Exploring how buffering strategies and bridging strategies interact also could impact SCRM. For example, Srinivasan and Swink (2015) found that firms investing in integration capabilities (i.e., bridging) would not experience expected benefits unless they also invested in the capability to quickly and effectively implement (i.e., buffering) any plans, which were subsequently developed. Similarly, Srinivasan and Swink (2018) found that if firms do not have the flexibility to quickly implement decisions, then access to more information is wasted. They argue that flexibility (i.e., buffering) and visibility (i.e., bridging) complement each other. Further, Revilla and Saenz (2017) found that firms engaging in both internal (i.e., buffering) and external (i.e., bridging) strategies had the best firm performance.

Although we find that SCRM improves overall firm performance, it is a broad metric. We need to understand which specific dimensions of buffering strategies or bridging strategies are associated with specific dimensions of performance. For example, buffering strategies' efforts to gain independence from suppliers through manufacturing flexibility such as quality improvement, inventory reduction, and faster cycle times and lead times can have financial benefits such as cost reduction and shorter cash cycles (Gerwin, 1993; Hayes & Wheelwright, 1984; Skinner, 1985). Alternatives such as excess capacity, financial savings, and the ability to re-route shipments can be enacted without disrupting marketing objectives such as pursuing new markets (Christopher & Holweg, 2011; Manuj & Mentzer, 2008b; Sheffi & Rice, 2005; Talluri et al., 2013). Similarly, bridging strategies' close relationships and information exchange could provide access to new technologies, which could be used for innovation objectives such as new product introduction.

Finally, how does culture result in varying effects? Although clustering by regions is rather broad, the results illuminate areas for more nuanced research. Why are bridging strategies' effect in Confucian Asia and Germanic Europe greater than in Anglo? How do cultural differences interact with missing potential moderators such as tier level? For example, do firms in some cultures conduct more or less offshoring of raw materials and components that might moderate the effectiveness of bridging strategies? What are the ways to overcome these differences? Are these fixed effects or are they changing over time?

Study Designs. Our meta-analysis reveals the confounding state of the SCRM field and points toward future work, which can provide consensus. The insufficient theoretical explication and lack of empirical

evidence create inconsistent use of constructs. This, in turn, causes researchers to utilize various measures due to the lack of consensus surrounding constructs. Ambiguity results in confusing and contradictory outcomes that lack the capability to advance theory (Flynn & Flynn, 2004). Operationalizations of constructs should, therefore, converge to enable comparison and advancement of theory. Indeed, this is one of the benefits of meta-analysis (Hunter & Schmidt, 2004).

The most common operationalization of supply chain disruption is the self-reported impact on six cross-functional metrics: prices, operational efficiency, quality, responsiveness, reliability, and sales (Bode et al., 2011). This is a parsimonious and holistic measure of outcomes, which should continue to be used. Supply chain disruption has been described with constructs such as duration, dichotomous severity, spread to other firms, and close calls. Habermann et al., 2015, who measure close calls, is the only study we found that begins to measure problem prevention. It is important for future work to continue to identify the successful prevention of problems in order to increase recognition of SCRM as strategic. The frequency of occurrence has been measured on a scale ranging from never to frequently (Zsidisin & Wagner, 2010). Measures of time in future work should move away from subjective self-reports (i.e., slow to fast) to more objective measures (i.e., hours, days, weeks).

One of the keys to buffering strategies is safety stock, which frequently measures perceptions of utilization. Zsidisin and Wagner (2010) include perceived degree of use of safety stock as an item in a scale for redundancy practices. In his four-item scale of safety stock, Park (2011) captures the perceived objective for holding inventory. Likert scales of perceptions of utilization, however, are subjective. Future work should utilize more objective measures to enable comparison across studies.

Some work has begun to measure more objective inventory levels (Brandon-Jones et al., 2015; Habermann et al., 2015). Most studies take the perspective of the focal firm. Brandon-Jones and colleagues, however, provide a very nuanced and parsimonious measure of the supply chain network perspective. They measure inventory levels of plants and suppliers separately. Due to the interdependencies in supply chain networks, this is an important operationalization that should be utilized in future research.

Supply chain resilience is an important SCRM construct, with several overlapping meanings. The broadest operationalizations define supply chain resilience as an outcome that includes both the readiness to prevent as well as recover from a disruption (Golgeci & Ponomarov, 2013; Park, 2011; Voss & Williams, 2013). A more narrow operationalization is a

dynamic capability enacted in the face of a supply chain disruption (Ambulkar et al., 2015; Brusset & Teller, 2017; Kwak, Seo, & Mason, 2018). The emphasis is quickly returning to normal or a more desirable state. These more narrow definitions separately attribute readiness and prevention to robustness (Wieland & Wallenburg, 2012). These differences are not minor. Given the centrality of this construct within the context of SCRM, consensus of the definition is necessary in order to consistently operationalize supply chain resilience. Only then we will be able to fully understand the construct and its relationship with other constructs.

We did not find a longitudinal study that met our sample criteria. This is an important gap, because a quick response time is important in reducing the impact of a supply chain disruption (Bode & Macdonald, 2017; Wagner & Bode, 2009). Actual response time was not measured, but perceptions of response time were measured with a Likert scale of 1 (slow) to 5 (fast) (Bode and Macdonald, 2017). More objective measures of time would enable unbiased comparisons.

Also, utilizing a longitudinal methodology to understand how SCRM evolves over time might help firms improve SCRM. We did not find a study that captured more than a snapshot of time. Research has shown that firms frequently initiate SCRM after experiencing a disruptive event (Bode et al., 2011), but we know little about how that initial experience impacts SCRM in the long term. Future work should examine how firms learn and adapt SCRM in response to supply chain disruptions.

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APPENDIX A PAPERS USED IN META-ANALYSIS

Author	Publication	Sample Size	Method	Level of Analysis	Globe Cluster	Theory
Ambulkar et al. (2015)	1	199	0	3	2	
Bode and Macdonald (2017)	1	438	0	3	0,2,3	6
Brandon-Jones et al. (2015)	1	264	0	2	0	6
Brandon-Jones et al. (2014)	1	264	0	2	0	4
Brusset and Teller (2017)	1	171	0	3	3	5
Cantor et al. (2014)	1	217	1	1	0	7
Chaudhuri, Boer and Taran (2018)	1	342	0	2	0	8
Chowdhury and Quaddus (2017)	1	296	0	3	1	5
Ellinger et al. (2015)	2	326	0	3	2	1
Gölgeci and Ponomarov (2013)	2	114	1	3	2	5
Habermann et al. (2015)	1	108	0	3	0	10
Kern et al. (2012)	1	162	0	3	0	
Kwak et al. (2018)	1	174	0	3	2	4
Lee and Rha (2016)	2	316	0	3	2	4
Liu and Lee (2018)	1	161	0	3	1	4
Mandal (2017)	2	207	0	3	2	4
Merschmann and Thonemann (2011)	1	85	0	3	2	9
Park (2011)	3	163	0	3	2	4,9
Riley (2013)	3	206	0	3	2	12
Shao (2013)	2	221	0	3	2	
Su (2011)	3	147	0	3	2	9
Viswanathan (2011)	3	153	0	3	2	13
Voss and Williams (2013)	1	338	0	3	2	3
Wieland and Wallenburg (2012)	1	268	0	3	0	
Wieland and Wallenburg (2013)	1	270	0	3	0	3,4,15
Zsidisin and Wagner (2010)	1	296	0	2,3	2	

Publication: 1-Top journal list outlined by Mackelprang and Nair (2010); 2- ABS Academic Journal Guide Peer-Reviewed Journal; 3-Dissertation

Method: 0-Field/Survey; 1-vignette survey mix

Level of analysis: 1-individual; 2-unit; 3-firm

Globe: 0-Anglo; 1-Confucian Asia; 2-Germanic Europe; 3-Latin Europe

Theory: 1-Knowledge-Based View; 2-Technology Acceptance Model; 3-Normal Accident Theory; 4-Resource Based View; 5-Dynamic Capability Theory; 6-Information Processing Theory; 7-Regulatory Focus Theory; 8-Agency Theory; 9-Contingency Theory; 10-Cluster Theory; 11-Convergence Theory; 12-High Reliability Theory; 13-Theory of Planned Behavior; 14-Strategic Choice Theory; 15-Resource Dependency Theory

APPENDIX B

DEFINITIONS OF CONSTRUCTS AND AGGREGATED ITEMS

Higher and Lower-level Constructs	Definition	Representative Measures
1. Supply chain risk management	"The identification, assessment, treatment, and monitoring of supply chain risks, with the aid of the internal implementation of tools, techniques and strategies and of external coordination and collaboration with supply chain members so as to reduce vulnerability and ensure continuity coupled with profitability, leading to competitive advantage" (Fan & Stevenson, 2018, p. 7).	Cantor et al. (2014); Ellinger et al. (2015)
Disruption impact (reverse coded)	Immediate and long-term detrimental consequences to operations and performance of the affected firm due to supply chain disruption (Blackhurst, Dunn, & Craighead, 2011).	Ambulkar et al. (2015); Bode and Macdonald (2017)
Disruption occurrence (reverse coded)	The frequency of supply chain disruptions (Zsidisin & Wagner, 2010).	Brandon-Jones et al. (2015)
Resilience	"The adaptive capability of the supply chain to prepare for unexpected events, respond to disruptions, and recover from them by maintaining continuity of operations at the desired level of connectedness and control over structure and function" (Ponomarev & Holcomb, 2009, p. 131).	Ambulkar et al. (2015); Chowdhury and Quaddus (2017)
Robustness	"The ability of the supply chain to maintain its function despite internal or external disruptions" (Brandon-Jones et al., 2014, p. 58).	Wieland and Wallenburg (2012); Brandon-Jones et al. (2014)
2. Buffering	Strategies to gain stability by establishing safeguards which minimize a firm's exposure to disturbances (Bode et al., 2011; Mishra et al., 2016).	All from aggregated items
Alternatives	The ability of a firm to quickly and effectively innovate, reconfigure, realign, and reorganize the supply chain in response to external uncertainties, thereby allowing the firm to provide solutions for customer needs and a superior competitive position (Ambulkar et al., 2015; Lee, Lee, & Schniederjans, 2011; Swafford, Ghosh, & Murthy, 2006). Includes measures of supply chain agility, supply chain flexibility, and supply chain reconfiguration.	Lee and Rha (2016); Kwak et al. (2018)
Independent	The ability to adapt to uncertainties through self-sufficient production sites possessing dedicated resources which specialize in products and technologies enabling versatility of production volume and mix (Flynn & Flynn, 1999; Skipper & Hanna, 2009). Includes measures of manufacturing flexibility and customer responsiveness.	Brusset and Teller (2017); Chaudhuri et al. (2018)
Protective	Increased resources and internal structures to secure and insulate the firm from external uncertainties which cannot be eliminated (Fennell & Alexander, 1987;	Zsidisin and Wagner (2010); Ambulkar et al. (2015)

(continued)

APPENDIX B (continued)

Higher and Lower-level Constructs	Definition	Representative Measures
3. Bridging	Meznar & Nigh, 1995; Voss & Williams, 2013). Includes measures of risk management infrastructure, security, and slack resources.	All from aggregated items
Cooperation	Strategies to manage uncertainty by creating linkages with supply chain network partners to increase influence over them (Bode et al., 2011; Mishra et al., 2016). A strategic set of methods and tools used to align, coordinate, and manage interdependence between organizations, people, and processes that interact to execute supply chain objectives (Mentzer et al., 2001; Xu & Beamon, 2006). Includes measures of collaboration and shared vision.	Park (2011); Wieland and Wallenburg (2013)
Information sharing	A strategic set of activities, tools, and methods for the exchange of information, coordination of functions, and alignment of processes to facilitate the synchronized flow of goods and services between supply chain network partners and share risks and rewards (Ellinger et al., 2015; Mackelprang et al., 2014). Includes measures of demand management, external capabilities, internal and external integration, IT connectivity, knowledge management, and visibility.	Wieland and Wallenburg (2013); Ellinger et al. (2015)
Joint activities	Strategically working together with supply chain network partners to plan, solve problems, and make decisions including performance evaluation, supplier development, continuous improvement projects, new product development, and risk management mechanisms (Mentzer et al., 2001; Spekman, Salmond, & Lambe, 1997; Stank, Keller, & Daugherty, 2001; Taran, Boer, & Lindgren, 2015).	Cantor et al. (2014); Bode and Macdonald (2017)
Monitoring	Methods and measures scanning the environment to detect and assess risk and sense changes, thereby increasing the likelihood of noticing early warning signals and form a quicker response (Bode & Macdonald, 2017).	Kern et al. (2012); Bode and Macdonald (2017)
4. Firm performance	The successful execution or accomplishment of work comprehensively operationalized with market, operations, and accounting criteria (Golobic & Smith, 2013).	Voss and Williams (2013); Lee and Rha (2016)
Competitive advantage	A unique position achieved relative to the competition in regard to superior efficiency, effectiveness, differentiation, and reputation (Hofer & Schendel, 1978; Kwak et al., 2018).	Kwak et al. (2018)
Cost performance	A condition of maximum output provided by worker efficiency, quality control, and waste reduction (Chowdhury & Quaddus, 2017; Färe & Lovell, 1978).	Chowdhury and Quaddus (2017)
Financial performance	Profit analyzed through the internal rate of return on assets (ROA) and return on investment (ROI) (Cantor et al., 2014; Tan, 2002).	Cantor et al., (2014)

(continued)

APPENDIX B (continued)

Higher and Lower-level Constructs	Definition	Representative Measures
Market performance	An assessment of a firm's ability to compete in the marketplace by their proportion of sales in the industry served (Buzzell, Gale, & Sultan, 1975).	Cantor et al. (2014); Chowdhury and Quaddus (2017)